Hoang Nam Dang

6namdang@gmail.com |LinkedIn | GitHub | Vietnamese citizen

Education

Gannon University – Bachelor of Science in Computer Science
GPA: 4.0/4.0
Coursework: Calculus 1 and 2, Data Structure and Algorithms, Numerical and Statistical Analysis, Distributed Programming, Mobile App Development, Object Oriented Programming, Operating System.

Skills

Programming Language: Python, Java, C#, JavaScript, C/C++, Go, SQL, NoSQL
Frameworks: .NET Core, FastAPI, Springboot, Django, Gin
Databases & Messaging: MySQL, PostgreSQL, MongoDB, InfluxDB, Redis, RabbitMQ
Tools & Platforms: Docker, Git, Vim, Linux, AWS, Azure

Experience

Software Engineer, E3 Company – Erie, PA **December 2024 – May 2025**

- Architected enterprise-scale IoT pipeline serving 10K+ devices with 99.9% uptime using microservices architecture (RabbitMQ, InfluxDB, MongoDB) on AWS EC2, reducing alert processing latency by 85%.
- Delivered 15x performance optimization for MongoDB alert queries under 20ms through strategic indexing and schema redesign, enabling real-time alerting for 50M+ daily events
- Built high-performance backend services in Java/JavaScript handling 1M+ API calls/day with Redis caching layer achieving sub 5ms response times, supporting 5x user growth.
- Led full-stack development of mission-critical alert management system using MUI/Redux frontend, serving enterprise clients with $2M+ annual contracts.

Machine Learning Research Assistant, Gannon University **Oct 2024 – Now**

- Researched on 2 different projects: International Students Admission Prediction advised by Dr. Kefei Wang and Intrusion Detection in Cloud Security – advised by Dr. Ronny Bazan
- Developed state-of-the-art models for time-series dataset to accurately predict non-matriculated students.
- Developed an economic dispatch, time-sensitive model for on-going traffic in AWS infrastructure with serverless lambda function.

VR Application Developer, Gannon University **Oct 2024 – May 2025**

- Engineered interactive 3D welding lathe simulator in Unity (C#) integrating Blender-modeled assets and NVIDIA PhysX physics, enabling ISM engineers to practice welding in VR and reduce material waste and safety hazards.
- Developed VR training application for industrial partner (ISM), representing manufacturing scenarios and lowering training costs by minimizing material consumption and on-site risks.
- Advisor: Professor Ohu Ikechukwu

Smart Manufacturing Research Assistant, Gannon University **Jan 2025 – July 2025**

- Collected and preprocessed 15000+ videos data from K1C machine, and automated labeling via Label Studio.
- Established, trained, and tested machine learning models for manufacturing applications; analyzed experimental results and assisted in paper drafting.
- Coordinated experiment sessions by communicating with principal investigators and participants, ensuring timely task completion and adherence to research protocols.
- Advisor: Professor Longfei Zhou

Projects

Interview Grader *(Sponsored by Lockheed Martin for Senior Capstone)* *500 Hours*

- Integrated AWS S3 (with AWS-SDK) for secure video storage and deployed optimized TensorFlow.js inference

(MediaPipe BlazeFace, MoveNet, Oarriaga's CNN) alongside a fine-tuned SmolVLM-BoEM on the BoLD dataset via a HuggingFace Serverless endpoint.

- Full-stack, real-time multimodal emotion recognition and pose detection system using Next.js and Supabase Auth to evaluate candidate nonverbal cues.

MNIST Hyper-param Visualizer – *Pytorch, Next.Js, C++, SQL*

- Full-stack web application that performed hyperparameter tuning (learning rate, batch size, optimizer) on MNIST classification in real-time, logging real-time loss/accuracy, visualizing metrics, and persisting model checkpoints/configurations for reproducibility.

Quantitative Option Trading Platform– *Python, Streamlit*

- Applied Monte Carlo simulation and volatility forecasting to optimize strike selection, visualize payoff diagrams, and compute Value-at-Risk (VaR) for portfolio hedging and performance attribution
- Developed a quantitative analytics tool for Iron Condor strategies, integrating real-time Yahoo Finance time series data, Black-Scholes option pricing, and Greeks (Delta, Gamma, Theta, Vega, Rho) to generate interactive P&L and risk models.

Research Papers

Interpretable ML for Transfer Student Success Prediction | First Author | ETLTC-ICES 2026

J.J. Durantz Research Award Recipient- Gannon University

- Developed Lasso Regression with L1 regularization ($R^2 = 0.334$), supported by an innovative composite feature selection methodology integrating SHAP values, Mutual Information, Random Forest importance, Recursive Feature Elimination (RFE), and permutation importance.

Novel approach for cost-effective Serverless Firewall using Machine Learning – IEEE Conference (In progress)

- In progress with Dr. Ronny Bazan and Master Md.Anisur